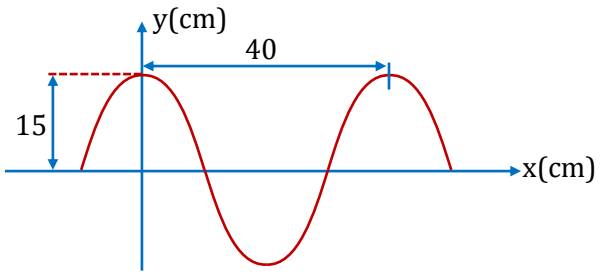
EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Equation of Travelling Wave

1. A sinusoidal wave travelling in the positive x direction has an amplitude of 15 cm, wavelength 40 cm and frequency 8 Hz. The vertical displacement of the medium at $t = 0$ and $x = 0$ is also 15 cm, as shown.



Find the angular wave number, angular frequency, speed of the wave and phase constant ϕ.

(A) $\frac{\pi}{20}, 16, 320, \frac{\pi}{2}$ (B) $\frac{\pi}{20}, 8, 160, \frac{\pi}{4}$ (C) $\frac{\pi}{10}, 16, 320, \frac{\pi}{4}$ (D) $\frac{\pi}{20}, 16, 320, \frac{\pi}{4}$

PWV341

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2. A plane progressive wave is represented by the equation $y = 0.25 \cos(2\pi t - 2\pi x)$. The equation of a wave is with double the amplitude and half frequency but travelling in the opposite direction will be

(A) $y = 0.5 \cos(\pi t - \pi x)$ (B) $y = 0.5 \cos(2\pi t + 2\pi x)$

(C) $y = 0.25 \cos(\pi t + 2\pi x)$ (D) $y = 0.5 \cos(\pi t + \pi x)$

PWV083

3. The equation of transverse wave is $y = a \cos 2\pi\left(\nu t - \frac{x}{\lambda}\right)$. The maximum particle velocity is twice of wave velocity if λ is-

(A) $\pi\frac{a}{2}$ (B) $2a$ (C) a (D) πa

PWV084

4. A travelling pulse is given by $f(x, t) = A \exp\left(\frac{2abxt - a^2x^2 - b^2t^2}{c^2}\right)$, where A, a, b, and c are positive constants of appropriate dimensions. The speed of the pulse is :

(A) $\frac{b}{a}$ (B) $\frac{2b}{a}$ (C) $\frac{cb}{a}$ (D) $\frac{b}{2a}$

PWV086

For More Join: @IITJEE_Advanced

5. A transverse sinusoidal wave moves along a string in the positive x direction. In figure (I) displacement of particle at P as a function of time is given and in figure (II) at a particular time t-the snap shot of wave is shown. The wave velocity (cm/s) and velocity of particle at P (cm/s) will be:

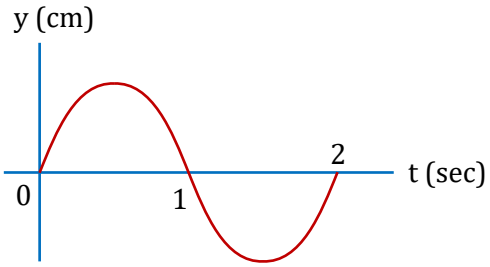


Figure-I

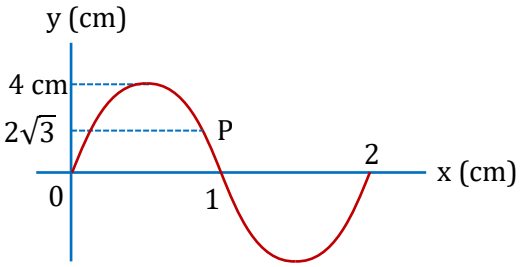


Figure-II

(A) $1\,\hat{i},-3\pi\,\hat{j}$ (B) $1\,\hat{i},3\pi\,\hat{j}$ (C) $1\,\hat{i},2\pi\,\hat{j}$ (D) $-1\,\hat{i},-2\pi\,\hat{j}$

PWV093

6. A wave propagates in a string in the positive x-direction with velocity v. The shape of the string at $t = t_0$ is given by $f(x, t_0) = A\sin\left(\frac{x^2}{a^2}\right)$: then, the wave equation at any instant t is given by

(A) $g(x, t) = A\sin\frac{\left[x-v(t-t_0)\right]^2}{a^2}$ (B) $g(x,t)=A\sin\frac{\left[x+v(t-t_0)\right]^2}{a^2}$

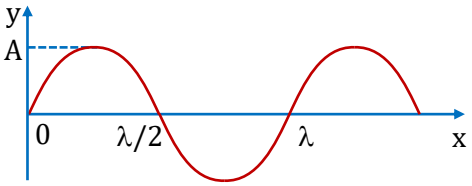
(C) $g(x,t)=A\sin\frac{\left[x-v(t+t_0)\right]^2}{a^2}$ (D) $g(x,t)=A\sin\frac{\left[x+v(t+t_0)\right]^2}{a^2}$

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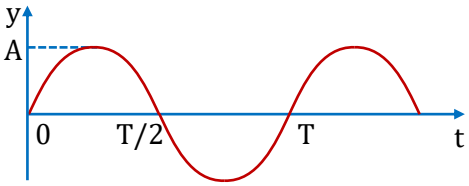
PWV096

7. Here given figure (i) shows snapshot at $t = T/4$ and figure (ii) shows motion of particle at $x = \lambda/4$. Then the possible equations of the wave will be $\left(\text{Here } T = \frac{2\pi}{\omega}\right)$:-

(i)



(ii)



(A) $y = A\sin\left(\omega t + kx - \frac{\pi}{2}\right)$

(B) $y = A\sin\left(\omega t - kx + \frac{\pi}{2}\right)$

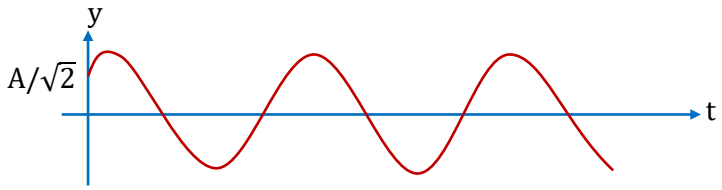
(C) Both $y = A\cos\left(\omega t + kx - \frac{\pi}{2}\right)$ and $y = A\cos\left(\omega t - kx + \frac{\pi}{2}\right)$

(D) Both $y = A\sin\left(\omega t + kx - \frac{\pi}{2}\right)$ and $y = A\sin\left(\omega t - kx + \frac{\pi}{2}\right)$

PWV098

For More Join: @IITJEE_Advanced

8. This is the graph of particle present at $x = \lambda/2$. Find the equation of wave travelling along $-x$ axis ?



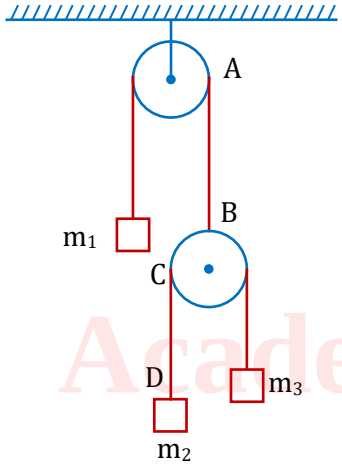
(A) $A \sin\left[\omega t - kx - \frac{3\pi}{4}\right]$ (B) $A \sin\left[\omega t + kx - \frac{3\pi}{4}\right]$

(C) $\frac{A}{\sqrt{2}} \sin\left[\omega t + kx - \frac{3\pi}{4}\right]$ (D) $\frac{A}{\sqrt{2}} \sin\left[\omega t - kx - \frac{3\pi}{4}\right]$

PWV100

Speed of Travelling Wave

9. Both the strings, shown in figure, are made of same material and have same cross-section. The pulleys are light. The wave speed of a transverse wave in the string AB is v_1 and in CD it is v_2. The v_1/v_2 is

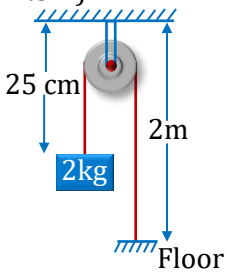


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(A) 1 (B) 2 (C) $\sqrt{2}$ (D) $\frac{1}{\sqrt{2}}$

PWV342

10. In the given figure the string has mass 4.5 g. Find the time taken by a transverse pulse produced at the floor to reach the pulley. ($g = 10\ ms^{-2}$).



(A) 0.015 sec (B) 0.02 sec (C) 0.035 sec (D) 0.045 sec

PWV107

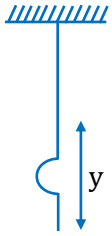
11. A rope AB of length L placed on a smooth horizontal surface and pulled by a horizontal force F at end B. The mass of rope is m. If a transverse wave travels from A to B in time T, the value of $\frac{9T^2F}{mL}$ is

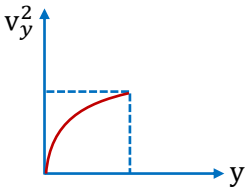
(A) 24 (B) 36 (C) 48 (D) 65

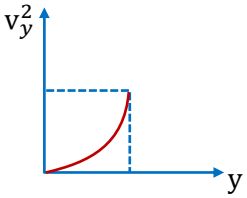
PWV109

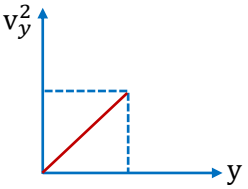
For More Join: @IITJEE_Advanced

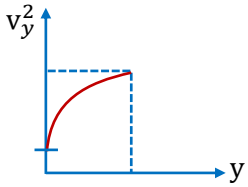
12. A non-uniform rope of length ℓ hangs from a ceiling. Mass per unit length of rope (μ) changes as $\mu = \mu_0 e^{y}$, where y is the distance along the string from its lowest point. Then graph between square of velocity of wave and y will be best represented as:



(A) 

(B) 

(C) 

(D) 

PWV238

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Energy and Power in Travelling Wave

13. A wave moving with constant speed on a uniform string passes the point $x = 0$ with amplitude A_0, angular frequency ω_0 and average rate of energy transfer P_0. As the wave travels down the string it gradually loses energy and at the point $x = \lambda$, the average rate of energy transfer becomes $\frac{P_0}{2}$.

At the point $x = \lambda$, angular frequency and amplitude are respectively:

(A) ω_0 and $\frac{A_0}{\sqrt{2}}$

(B) $\frac{\omega_0}{\sqrt{2}}$ and A_0

(C) Less than ω_0 and A_0

(D) $\omega_0/\sqrt{2}$ and $A_0/\sqrt{2}$

PWV145

14. A transverse wave is passing through a stretched string with a speed of 20 m/s. The tension in the string is 20 N. At a certain point P on the string, it is observed that energy is being transferred at a rate of $40mW$ at a given instant. Find the speed of point P.

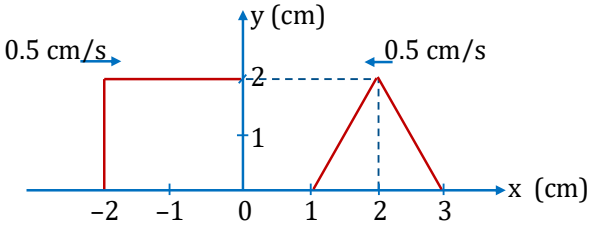
(A) 40 cm/s (B) 20 cm/s (C) 2 mm/s (D) 20 mm/s

PWV148

For More Join: @IITJEE_Advanced

Superposition of Waves

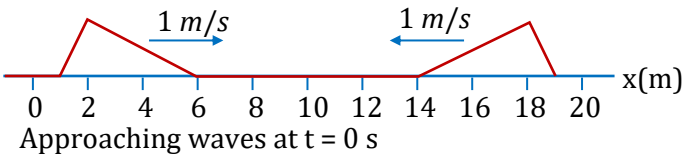
15. Figure shows a rectangular pulse and a triangular pulse approaching each other along *x*-axis. The pulse speed is 0.5 *cm/s*. What is the resultant displacement of medium particles due to superposition of waves at $x = 0.25$ *cm* and $t = 1$ *sec*.?



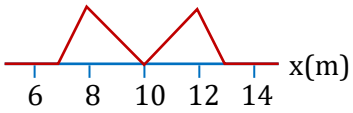
(A) 1 *cm* (B) 2 *cm* (C) 2.5 *cm* (D) 3 *cm*

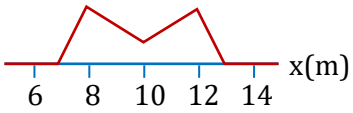
PWV113

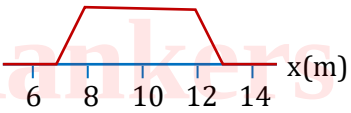
16. Two pulses on a string approach each other at speeds of 1 *m/s*. What is the shape of the string at $t = 6s$?

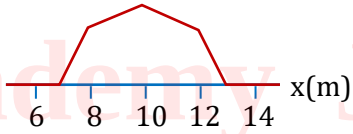


Approaching waves at t = 0 s

(A) 

(B) 

(C) 

(D) 

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PWV114

17. Two waves having equations

$$x_1 = a \sin(\omega t + \phi_1), \quad x_2 = a \sin(\omega t + \phi_2)$$

If in the resultant wave the frequency and amplitude remain equal to those of superimposing waves. Then phase difference between them is

(A) $\frac{\pi}{6}$ (B) $\frac{2\pi}{3}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{3}$

PWV116

18. Two waves having the intensities in the ratio of 9 : 1 produce interference. The ratio of maximum to the minimum intensity, is equal to

(A) 2 : 1 (B) 4 : 1 (C) 9 : 1 (D) 10 : 8

PWV118

19. Five waveforms moving with equal speeds on the *x*-axis

$y_1 = 8 \sin(\omega t + kx)$; $y_2 = 6 \sin(\omega t + \frac{\pi}{2} + kx)$; $y_3 = 4 \sin(\omega t + \pi + kx)$; $y_4 = 2 \sin(\omega t + \frac{3\pi}{2} + kx)$;

$y_5 = 4\sqrt{2} \sin(\omega t - kx + \frac{\pi}{4})$ are superimposed on each other. The resulting wave is :

(A) $8\sqrt{2}\cos kx \sin(\omega t + \frac{\pi}{4})$ (B) $8\sqrt{2}\sin(\omega t - kx + \frac{\pi}{4})$

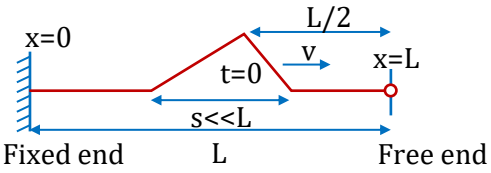
(C) $8\sqrt{2}\sin kx \cos(\omega t + \frac{\pi}{4})$ (D) $8 \sin(\omega t + kx)$

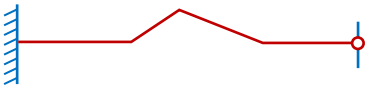
PWV124

For More Join: @IITJEE_Advanced

Reflection and Transmission of Waves

20. A small pulse travelling with speed v in a string is shown at $t = 0$, moving towards free end. Which of these is not **CORRECTLY** matched.



(i) $t = \frac{L}{v}$ (P) 

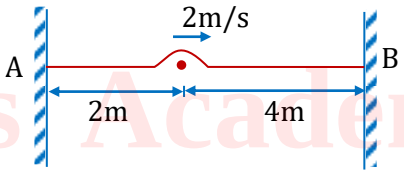
(ii) $t = \frac{2L}{v}$ (Q) 

(iii) $t = \frac{3L}{v}$ (R) 

(A) (i) (B) (ii) (C) (iii) (D) None of these

PWV120

21. A string is tied at two rigid supports. A pulse is generated on the string as shown in figure. Minimum time after which string will regain its shape: (Neglect the time during reflection)



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(A) 2 sec (B) 4 sec (C) 6 sec (D) None of these

PWV121

22. A wave is represented by the equation $y = 10 \sin 2\pi(100t - 0.02x) + 10 \sin 2\pi(100t + 0.02x)$. The maximum amplitude and loop length are respectively

(A) 20 units and 30 units (B) 20 units and 25 units

(C) 30 units and 20 units (D) 25 units and 20 units

PWV122

23. A composite string is made up by joining two strings of different masses per unit length μ and 4μ respectively. The composite string is under the same tension. A transverse wave pulse $Y = (6\ mm) \sin (5t + 40x)$, where t is in seconds and x in metres, is sent along the lighter string towards the joint. The joint is at $x = 0$. The equation of the wave pulse reflected from the joint is

(A) $(2mm) \sin(5t - 40x)$ (B) $(4mm) \sin(40x - 5t)$

(C) $-(2mm) \sin(5t - 40x)$ (D) $(2mm) \sin(5t - 10x)$

PWV125

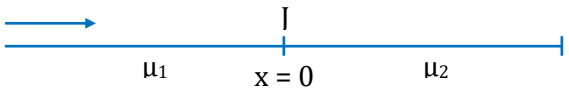
24. A wave of amplitude $10cm$ travelling on a string is reflected from denser string. Energy of transmitted wave is 64% of the incident wave. Amplitude of reflected wave is

(A) 6 *cm* (B) 6.4 *cm* (C) 8 *cm* (D) None of these

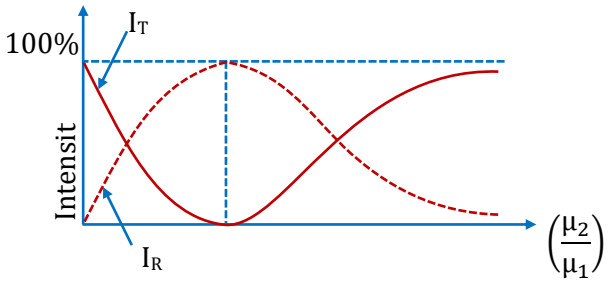
PWV126

For More Join: @IITJEE_Advanced

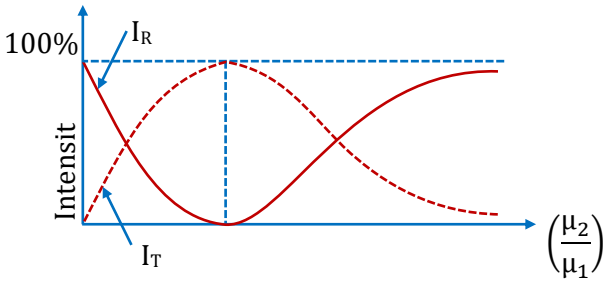
25. A transverse wave travelling in a string of linear mass density μ_1 is incident on junction point *J*. The intensities of reflected and transmitted wave both as a function of $\left(\frac{\mu_1}{\mu_2}\right)$ are plotted. The correct sketch is :



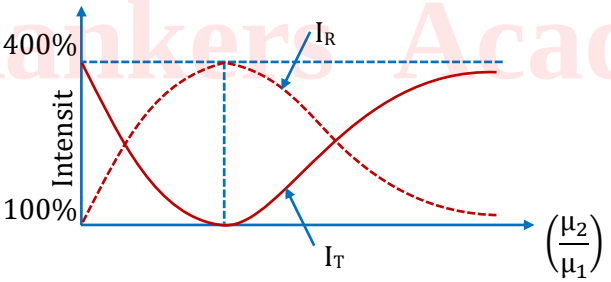
(A)



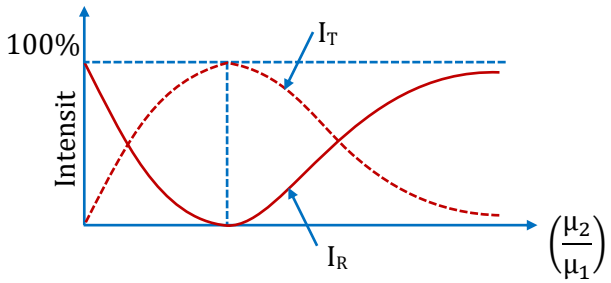
(B)



(C)



(D)



PWV241

Standing Waves

26. Which two of the given transverse waves will give stationary waves when get superimposed

$z_1 = a\cos(kx - \omega t)$ ...(A)

$z_2 = a\cos(kx + \omega t)$ ...(B)

$z_3 = a\cos(ky - \omega t)$ ...(C)

(A) *A* and *B* (B) *A* and *C* (C) *B* and *C* (D) Any two

PWV127

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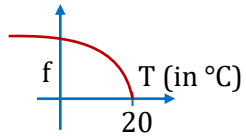
27. A standing wave having 3 nodes and 2 antinodes is formed between two atoms having a distance 1.21 $\mathring{A}$ between them. The wavelength of the standing wave is
(A) 1.21 $\mathring{A}$ (B) 2.42 $\mathring{A}$ (C) 6.05 $\mathring{A}$ (D) 3.63 $\mathring{A}$

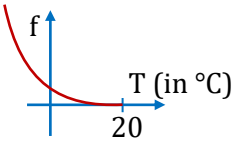
PWV132

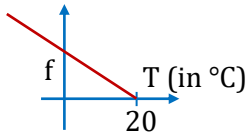
28. A standing wave is represented by $\boldsymbol{Y = A\sin(100t)\cos(0.01x)}$ where Y and A are in *millimeter*, t is in seconds and x is in *meter*. The velocity of wave is
(A) $10^4 m/s$ (B) $1 m/s$
(C) $10^{-4} m/s$ (D) Not derivable from above data

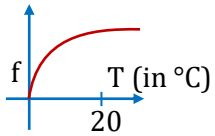
PWV135

29. A metal wire is clamped between two vertical walls. At 20°C the unstrained length of the wire is exactly equal to the separation between walls. If the temperature of the wire is decreased the graph between fundamental frequency (f) and temperature (T) of the wire is

(A) 

(B) 

(C) 

(D) 

PWV141

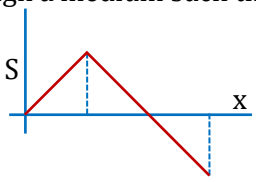
30. The ends of a stretched wire of length L are fixed at $x = 0$ and $x = L$. In one experiment, the displacement of the wire is $y_1 = A\sin(\pi x/L)\sin\omega t$ and energy is E_1 and in another experiment its displacement is $y_2 = A\sin(2\pi x/L)\sin 2\omega t$ and energy is E_2. Then
(A) $E_2 = E_1$ (B) $E_2 = 2E_1$ (C) $E_2 = 4E_1$ (D) $E_2 = 16E_1$

PWV152

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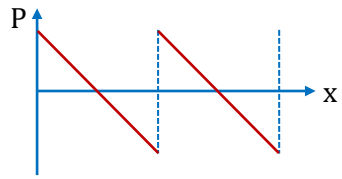
Displacement and Pressure Wave Equation

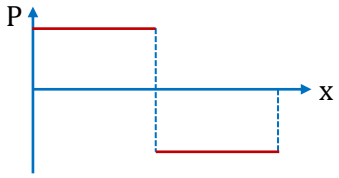
31. A sound wave is propagating through a medium such that displacement v/s x graph is

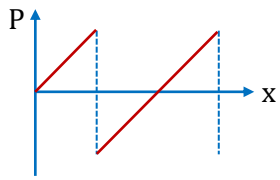


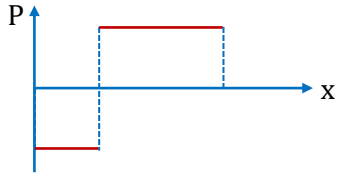
$\Delta p = -\beta\left(\frac{\delta s}{\delta x}\right) = \beta s_0 k\sin(kx - \omega t)$

Corresponding excess pressure v/s x graph will be

(A) 

(B) 

(C) 

(D) 

PWV155

For More Join: @IITJEE_Advanced

Properties of Sound

32. Mark the **INCORRECT** statement about longitudinal pressure wave.
(A) The density variation and the longitudinal velocity of the particles have their maxima and minima at the position of equilibrium of the particles.
(B) At the position of maximum displacement, the density is at the ambient (environment) value.
(C) Greatest compression occurs at the same time and place where the velocity of the particles is maximum and in direction of wave propagation.
(D) Greatest rarefaction occurs at the same time and place where the velocity of the particles is maximum and in direction of wave propagation.

PWV156

33. If the velocity of sound in hydrogen at $27°C$ is 1200 ms^{-1}, velocity of sound in oxygen at the same temperature will be
(A) 3740 ms^{-1} (B) 300 ms^{-1} (C) 1000 ms^{-1} (D) 2000 ms^{-1}

PWV157

34. It is found that an increase in pressure of 100 kPa causes a certain volume of water to decrease by 5×10^{-3} percent of its original volume. Then the speed of sound in the water is about (density of water $10^3\ kg/m^3$)
(A) 330 m/s (B) 1414 m/s (C) 1732 m/s (D) 2500 m/s

PWV158

35. A point source of sound is located somewhere along the x-axis. Experiments show that the same wave front simultaneously reaches listeners at $x = -8\ m$ and $x = +2.0\ m$.
A third listener is positioned along the positive y-axis. What is her y-coordinate (in m) if the same wave front reaches her at the same instant as it does the first two listeners?
(A) 4 (B) 3 (C) 2 (D) 5

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PWV159

36. At what temperature velocity of sound is double than that of at $0°C$
(A) 819 K (B) $819°C$ (C) $600°C$ (D) 600 K

PWV162

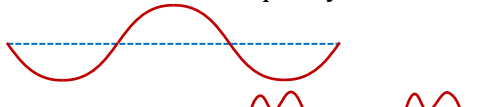
37. The speed of longitudinal wave is 100 times the speed of transverse wave in a taut brass wire. If the Young's modulus of brass is $1.0 \times 10^{11}\ N/m^2$, the stress in wire is
(A) $1.0 \times 10^7\ N/m^2$ (B) $1.0 \times 10^6\ N/m^2$ (C) $1.0 \times 10^5\ N/m^2$ (D) $1.0 \times 10^8\ N/m^2$

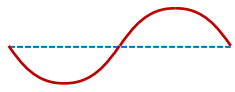
PWV164

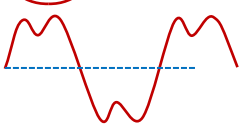
38. A is singing a note and at the same time B is also singing a note with $1/8^{th}$ the frequency of A. The energies of the two sounds are equal. The displacement amplitude of the note of B is:
(A) same as that of A
(B) twice that of A
(C) four times that of A
(D) eight times that of A

PWV166

39. A microphone is connected to an oscilloscope. The diagram shows the trace on the screen when the microphone receives a pure note. Which trace can be obtained when a musical instrument produces a note of the same pitch but of a different quality?



(A) 

(B) 

(C) 

(D) 

PWV167

For More Join: @IITJEE_Advanced

40. A sound absorber attenuates the sound level by 20 *dB*. The intensity decreases by a factor of:-
(A) 1000 (B) 10000 (C) 10 (D) 100

PWV171

Interference of Waves

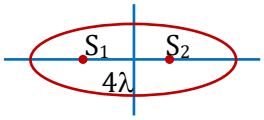
41. Which of the following is the equation of a spherical wave?
(A) $S = S_0 \sin(Kx - \omega t)$ (B) $S = S_0 \cos(Kx - \omega t)$
(C) $S = (S_0/x) \sin(\omega t - Kx)$ (D) $S = (S_0/x^2) \sin(\omega t - Kx)$

PWV169

42. The ratio of intensities between two coherent sound sources is 4 : 1. The difference of loudness in *dB* between maximum and minimum intensities when they interfere in space is:
(A) 10 log 2 (B) 20 log 3 (C) 10 log 3 (D) 20 log 2

PWV176

43. S_1, S_2 are two coherent sources of sound located along x – axis separated by 4λ where λ is wavelength of sound emitted by them. Number of maxima located on the elliptical boundary around it will be :

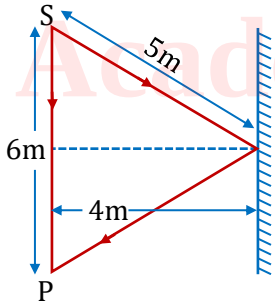


(A) 16 (B) 12 (C) 8 (D) 4

PWV177

44. A person standing at a distance of 6*m* from a source of sound receives sound wave in two ways, one directly from the source and other after reflection from a rigid boundary as shown in the figure. The maximum wavelength for which, the person will receive maximum sound intensity, is

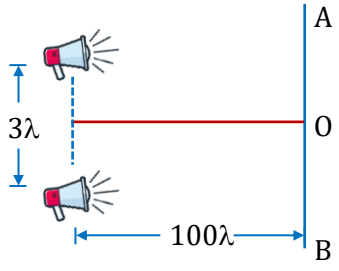
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(A) 4 *m* (B) $\frac{16}{3}$ *m* (C) 2 *m* (D) $\frac{8}{3}m$

PWV178

45. Two loudspeakers are emitting sound waves of wavelength λ with an initial phase difference of $\frac{\pi}{2}$.
At what minimum distance from *O* on the line *AB* will one hear a maxima?



(A) 25 λ (B) $\frac{100\lambda}{\sqrt{15}}$ (C) $\frac{25\lambda}{3}$ (D) 50 λ

PWV179

For More Join: @IITJEE_Advanced

Standing Waves and Beats

46. The fundamental frequency of a closed organ pipe is same as the first overtone frequency of an open organ pipe. If the length of open pipe is 50 *cm*, the length of closed pipe is:-

(A) 25 *cm* (B) 12.5 *cm* (C) 100 *cm* (D) 200 *cm*

PWV187

47. The second overtone of an open pipe *A* and a closed pipe *B* have the same frequencies. The ratio of fundamental frequency of *A* to the fundamental frequency of *B* is:

(A) 3 : 5 (B) 5 : 3 (C) 5 : 6 (D) 6 : 5

PWV188

48. A closed organ pipe is vibrating in its second overtone. The length of the pipe is 10*cm* and maximum amplitude of vibration of particles of the air in the pipe is 2*mm*. Then the amplitude of S.H.M. of the particles at 9*cm* from the open end is:

(A) $\sqrt{3}\,mm$ (B) $\sqrt{2}\,mm$ (C) $\frac{\sqrt{3}}{2}\,mm$ (D) None of these

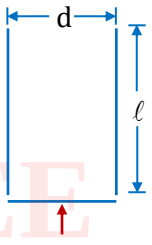
PWV199

49. A resonance tube is resonated with tuning fork of frequency 256 *Hz*. If the length of first and second resonating air columns are 32 *cm* and 100 *cm*, then end correction will be

(A) 1 *cm* (B) 2 *cm* (C) 4 *cm* (D) 6 *cm*

PWV200

50. A tube of diameter d and of length ℓ unit is open at both the ends. Its fundamental frequency of resonance is found to be f_1. The velocity of sound in air is 330 m/sec. One end of tube is now closed. The lowest frequency of resonance of tube is now f_2. Taking into consideration the end correction, $\frac{f_2}{f_1}$ is



(A) $\frac{(\ell+0.6\times d)}{(\ell+0.3\times d)}$ (B) $\frac{1}{2}\frac{(\ell+0.3\times d)}{(\ell+0.6\times d)}$ (C) $\frac{1}{2}\frac{(\ell+0.6\times d)}{(\ell+0.3\times d)}$ (D) $\frac{1}{2}\frac{(d+0.3\ell)}{(d+0.6\ell)}$

PWV201

51. A resonance tube is resonated with tuning fork of frequency 256 *Hz*. If the length of first and second resonating air columns are 32 *cm* and 100 *cm*, then end correction will be

(A) 1 *cm* (B) 2 *cm* (C) 4 *cm* (D) 6 *cm*

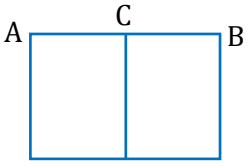
PWV343

52. Two organ pipes, each closed at one end, give 6 beats per sec, when emitting their fundamental nodes. If their lengths are in the ratio of 6 : 5, their fundamental frequencies (in *Hz*) are f_1 and f_2 then find the value of $f_1 + f_2$.

(A) 66 (B) 62 (C) 68 (D) 64

PWV206

53. *AB* is a cylinder of length 1.0 *m* filled with a thin flexible adibatic membrane *C* (see figure) at the middle and two other thin flexible adiabatic membranes *A* and *B* at the ends. The portions *AC* and *BC* contain hydrogen and oxygen gasses respectively at same pressure. Standing waves are setup in two columns such that membrane *A* acts as antinode and *B* as node (speed of sound in oxygen is 330 m/s) The frequency of these vibrations for which the membrane *C* is a node is :



(A) 330 *Hz* (B) 550 *Hz* (C) 1485 *Hz* (D) 660 *Hz*

PWV243

For More Join: @IITJEE_Advanced

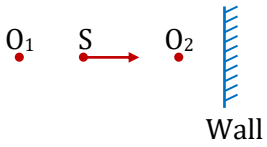
54. Three sound waves of equal amplitudes have frequencies (ν – 1), ν, (ν + 1). They superpose to give beats. The number of beats produced per second will be :-

(A) 2 (B) 1 (C) 4 (D) 3

PWV242

Doppler Effect

55. A source of sound producing sound of frequency 100 Hz is moving towards a wall with velocity 20 m/s as shown. If O_1 and O_2 are two stationary observers and beat frequencies heard by them are f_1 and f_2 respectively between sound directly from source and sound reflected from wall. Then:

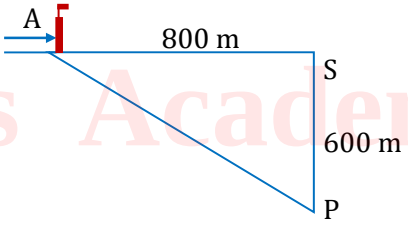


(Assume velocity of sound to be 330 m/s)

(A) $f_1 = 0\ Hz$ (B) $f_1 = f_2$ (C) $f_1 > f_2$ (D) $f_1 < f_2$

PWV212

56. A person P is 600 m away from the station when train is approaching station with 72 km/h, it blows a whistle of frequency 800 Hz when 800 m away from the station. Find the frequency heard by the person. Speed of sound = 340 ms^{-1}.



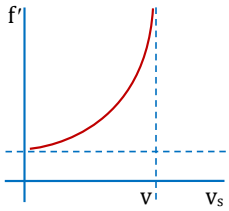
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(A) 800 Hz (B) 839.5 Hz (C) 829.5 Hz (D) 843.5 Hz

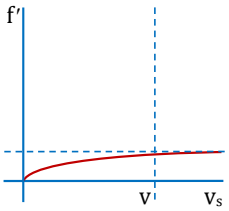
PWV214

57. Which of the following graphs represents a plot of the frequency f' of a sound wave perceived by a stationary observer versus the speed v_s of the source approaching towards him?

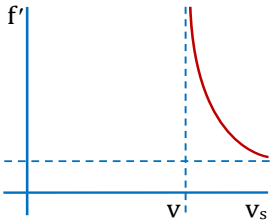
(A)



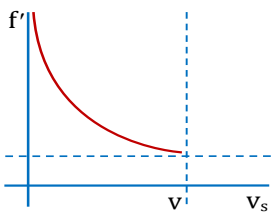
(B)



(C)



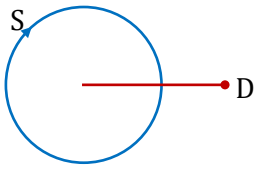
(D)



PWV216

For More Join: @IITJEE_Advanced

58. A source of sound moves along a circle of radius 2 m with constant angular velocity 40 radian/s. Frequency of the source is 300 Hz. A detector is kept of some distance from the circle in the same plane of the circle (as shown in figure). Which of the following is not the possible value of frequency registered by the detector ? (Speed of sound = 320 m/s)



(A) 250 Hz (B) 360 Hz (C) 396 Hz (D) 220 Hz

PWV221

59. A source of frequency n and an observer are moving on a straight line with velocities a and b respectively. If the source is ahead of the observer and if the medium is also moving in the direction of their motion with velocity c (the velocity of sound being v) then the apparent frequency of sound heard by the observer would be:

(A) $\left(\frac{v+c+b}{v+c+a}\right)n$ (B) $\left(\frac{v-c+b}{v-c+a}\right)n$ (C) $\left(\frac{v-c-b}{v-c-a}\right)n$ (D) $\left(\frac{v+c-b}{v+c-a}\right)n$

PWV222

60. The observed wavelength of light coming from a distant galaxy is found to be increased by 0.5% as compared with that coming from a terrestrial source. The galaxy is

(A) Stationary with respect to the earth

(B) Approaching the earth with velocity of light

(C) Receding from the earth with the velocity of light

(D) Receding from the earth with a velocity equal to $1.5 \times 10^6 m/s$

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PWV231

61. A star producing light of wavelength 6000 Å moves away from the earth with a speed of 5 km/sec. Due to Doppler effect the shift in wavelength will be ($c = 3 \times 10^8 m/\sec$)

(A) 0.1 Å (B) 0.05 Å (C) 0.2 Å (D) 1 Å

PWV232

62. If the shift of wavelength of light emitted by a star is towards violet, then this shows that star is

(A) Stationary (B) Moving towards earth

(C) Moving away from earth (D) Information is incomplete

PWV233

63. Assuming that universe is expanding, if the spectrum of light coming from a star which is going away from earth is tested, then in the wavelength of light

(A) There will be no change

(B) The spectrum will move to infrared region

(C) The spectrum will seems to shift to ultraviolet side

(D) None of the above

PWV234

64. A light source approaches the observer with velocity 0.8 c. The doppler shift for the light of wavelength 5500 $\AA$ is

(A) 4400 $\AA$ (B) 1833 $\AA$ (C) 3167 $\AA$ (D) 7333 $\AA$

PWV236

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EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. A wave equation which gives the displacement along the y direction is given by $y = 10^{-4} \sin(60t + 2x)$, where x and y are in meters and t is time in seconds. This represents a wave:

(A) travelling with a velocity of 30 ms^{-1} in the negative x direction.

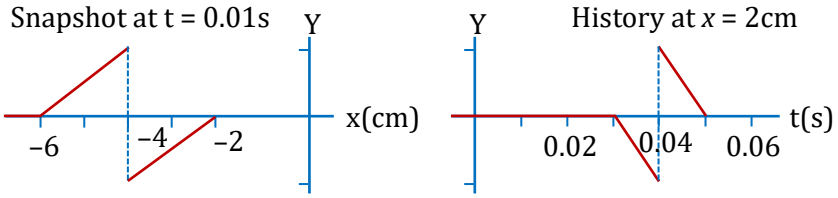
(B) of wavelength π m

(C) of frequency $30/\pi$ Hz

(D) of amplitude 10^{-4} m

PWV087

2. Figure shows a snapshot graph and a history graph for a wave pulse on a stretched string. They describe the same wave from two perspectives.



(A) The wave is travelling in positive x-direction.

(B) The wave is travelling in negative x-direction.

(C) The speed of the wave is 2 m/s.

(D) The peak is located at $x = -6$ cm at $t = 0$.

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PWV058

3. Consider a hypothetical wave pulse (at time $t = 0$) given by the following (y, x in meter)

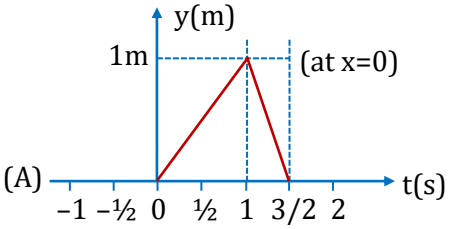
$y = 0, x < 0$

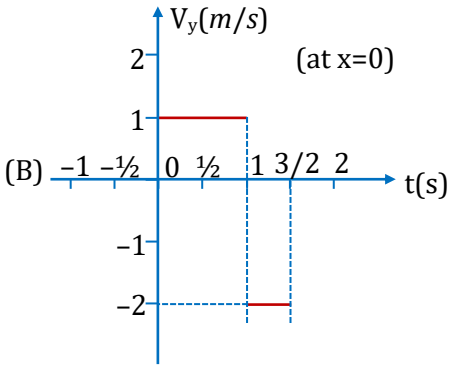
$y = x/2, 2 > x \geq 0$

$y = 3 - x, 3 \geq x \geq 2$

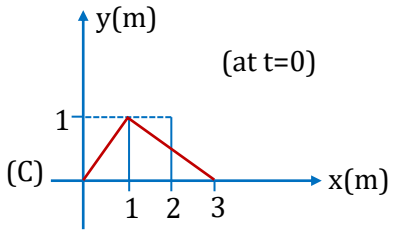
$y = 0, x > 3$

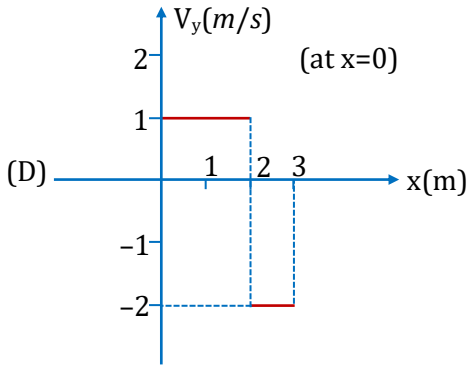
The pulse travels leftwards (negative x direction) at speed $v = 2$ m/s. Which of the following plots are correct? [V_y is the velocity of particle]





For More Join: @IITJEE_Advanced

(C)


(D)


PWV056

4. Two mechanical waves, $y_1 = 2 \sin 2\pi (50\, t - 2x)$ and $y_2 = 4 \sin 2\pi (ax + 100\, t)$ propagate in a medium with same speed.

(A) The ratio of their intensities is 1 : 16 (B) The ratio of their intensities is 1 : 4

(C) The value of 'a' is 4 units (D) The value of 'a' is 2 units

PWV146

5. An string has resonant frequencies given by 1001 *Hz* and 2639 *Hz*.

(A) If the string is fixed at one end only, 910 *Hz* can be a resonance frequency.

(B) If the string is fixed at one end only, 1911 *Hz* can be a resonance frequency.

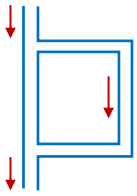
(C) If the string is fixed at both the ends, 364 *Hz* can be one of the resonant frequency.

(D) 1001 *Hz* is definitely not the fundamental frequency of the string.

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PWV134

6. A sound consists of four frequencies → 300 *Hz*, 900 *Hz*, 2400 *Hz* and 4500 *Hz*. A sound 'filter' is made by passing this sound through a bifurcated pipe as shown. The sound waves have to travel a distance of 50 *cm* more in the right branch-pipe than in the straight pipe. The speed of sound in air is 300 *m/s*. Then, which of the following frequencies will be almost completely muffled or "silenced" at the outlet?



(A) 300 *Hz* (B) 900 *Hz* (C) 2400 *Hz* (D) 4500 *Hz*

PWV055

7. Two tuning forks of frequency 250 *Hz* and 256 *Hz* produce beats. If a maximum of intensity is observed just now, after how much time the minimum is observed at the same place ?

(A) $\frac{1}{18}$ sec. (B) $\frac{1}{4}$ sec. (C) $\frac{1}{3}$ sec. (D) $\frac{1}{12}$ sec.

PWV053

For More Join: @IITJEE_Advanced

8. A car moves towards a hill with speed v_c. It blows a horn of frequency f which is heard by an observer following the car with speed v_0. The speed of sound in air is v.

(A) The wavelength of sound reaching the hill is $\frac{v}{f}$.

(B) The wavelength of sound reaching the hill is $\frac{v-v_c}{f}$.

(C) The beat frequency observed by the observer is $\left(\frac{v+v_o}{v-v_c}\right)f$.

(D) The beat frequency observed by the observer is $\frac{2v_c(v+v_o)f}{v^2-v_c^2}$.

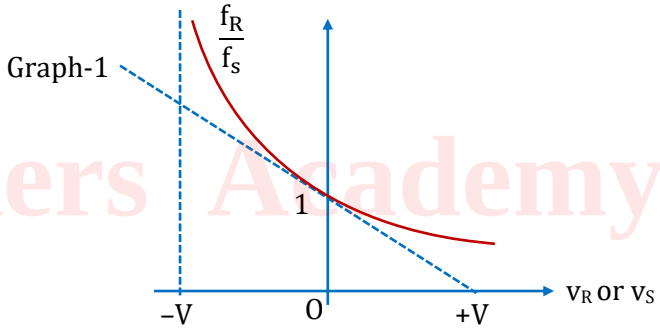
PWV054

9. Consider :

Case-1 : A moving source moving with velocity v_s and a stationary receiver.

Case-2 : A moving receiver moving with velocity v_R and a stationary source.

In either case f_R represents frequency heard by receiver and f_S represents frequency emitted by source. Figure shows graph of $\frac{f_R}{f_S}$ on y axis and v_s or v_R on x axis. Mark the **CORRECT** option (s)



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(A) Graph-1 corresponds to moving receiver and Graph-2 corresponds to moving source.

(B) Graph-2 corresponds to moving receiver and Graph-1 corresponds to moving source.

(C) Graphs to left of origin represents approaching listener and to right represents receding listener.

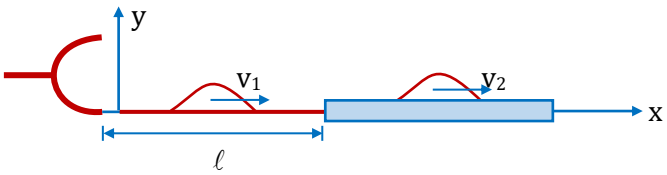
(D) Graphs to right of origin represents approaching source and to left represent receding source.

PWV244

COMPREHENSION TYPE QUESTIONS

PARAGRAPH FOR QUESTION NOS. 10 TO 12

A harmonic oscillator at $x = 0$, oscillates with a frequency $\frac{\omega}{2\pi}$ and amplitude a. It is generating waves at end of a thin string in which velocity of wave is v_1 and which is connected to another heavier string in which velocity of wave is v_2 as shown, length of first string is ℓ.



For More Join: @IITJEE_Advanced

10. If harmonic oscillator oscillates by an equation $y = a \sin \omega t$. The equation of incident wave in first string is

(A) $y = a \sin \omega \left(t - \frac{x}{v_1}\right)$ (B) $y = a \sin \omega \left(t + \frac{x}{v_1}\right)$

(C) $y = a \sin \left[\omega \left(t - \frac{x}{v_1}\right) + \pi\right]$ (D) $y = a \sin \left[\omega \left(t + \frac{x}{v_1}\right) + \pi\right]$

PWV128

11. Equation of transmitted wave in second string if its amplitude is a_t is

(A) $y = a_t \sin \omega \left(t - \frac{x}{v_2}\right)$ (B) $y = a_t \sin \omega \left(t - \frac{\ell}{v_1}\right)$

(C) $y = a_t \sin \omega \left(t - \frac{\ell}{v_1} - \frac{x-\ell}{v_2}\right)$ (D) $y = a_t \sin \omega \left(t - \frac{x}{v_2}\right)$

PWV129

12. Equation of reflected wave, if it is reflecting at the joint and amplitude of reflected wave is a_R

(A) $y = a_R \sin \omega \left(t - \frac{x}{v_2}\right)$ (B) $y = a_R \sin \left[\omega \left(t - \frac{\ell}{v_1} - \frac{\ell - x}{v_1}\right) + \pi\right]$

(C) $y = a_R \sin \left[\omega \left(t + \frac{x}{v_1}\right) + \pi\right]$ (D) $y = a_R \sin \left[\omega \left(t + \frac{2\ell + x}{v_1}\right) + \pi\right]$

PWV130

PARAGRAPH FOR QUESTION NO. 13 TO 15

A metallic rod of length $1m$ has one end free and other end rigidly clamped. Longitudinal stationary waves are set up in the rod in such a way that there are total six antinodes present along the rod. The amplitude of an antinode is 4×10^{-6} m. Young's modulus and density of the rod are 6.4×10^{10} N/m^2 and 4×10^3 Kg/m^3 respectively. Consider the free end to be at origin and at $t = 0$ particles at free end are at positive extreme.

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13. The equation describing displacements of particles about their mean positions is

(A) $s = 4\times10^{-6} \cos\left(\frac{11\pi}{2}x\right)\cos(22\pi\times10^3 t)$ (B) $s = 4\times10^{-6} \cos\left(\frac{11\pi}{2}x\right)\sin(22\pi\times10^3 t)$

(C) $s = 4\times10^{-6} \cos(5\pi x)\cos(20\pi\times10^3 t)$ (D) $s = 4\times10^{-6} \cos(5\pi x)\sin(20\pi\times10^3 t)$

PWV194

14. The equation describing stress developed in the rod is

(A) $140.8\pi\times10^4 \cos\left(\frac{11}{2}\pi x + \pi\right)\cos(22\pi\times10^3 t)$

(B) $140.8\pi\times10^4 \sin\left(\frac{11}{2}\pi x + \pi\right)\cos(22\pi\times10^3 t)$

(C) $128\pi\times10^4 \cos(5\pi x + \pi)\cos(20\pi\times10^3 t)$

(D) $128\pi\times10^4 \sin(5\pi x + \pi)\cos(20\pi\times10^3 t)$

PWV195

15. The magnitude of strain at midpoint of the rod at $t = 1$ sec is

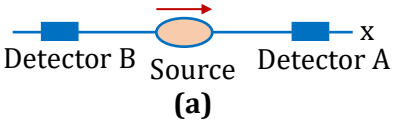
(A) $11\sqrt{3}\pi\times10^{-6}$ (B) $11\sqrt{2}\pi\times10^{-6}$ (C) $10\sqrt{3}\pi\times10^{-6}$ (D) $10\sqrt{2}\pi\times10^{-6}$

PWV196

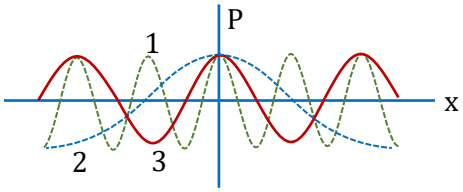
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PARAGRAPH FOR QUESTION NO. 16 TO 19

A source emitting a sound wave at a certain frequency moves with constant speed along an x-axis figure (a). The source moves directly towards a stationary detector A and directly away from another stationary detector B. The superimposed three plots of figure (b) indicate the pressure function $P(x)$ of the sound wave as measured by detector A, by detector B, and by someone (c) in the rest frame of the source.



(a)



(b)

16. Which of the following plot corresponds to the measurement done by detector A?

(A) 1 (B) 2

(C) 3 (D) These plots are not possible

PWV217

17. The plot corresponding to the measurement done by detector B is

(A) 1 (B) 2

(C) 3 (D) These plots are not possible

PWV218

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18. The plot corresponding to the measurement done by the detector C is

(A) 1 (B) 2

(C) 3 (D) These plots are not possible

PWV219

19. Now the source stops and begins to move along y-axis with same speed, the plot which corresponds to the measurement of B now is

(A) 1 (B) 2 (C) 3 (D) none of these

PWV220

PARAGRAPH FOR QUESTION NO. 20 AND 21

A thin string is held at one end and oscillated vertically, so that

$y(x = 0, t) = 8 \sin 4t\ [cm]$

Neglect the gravitational force. The string's density is $0.2\left[\frac{kg}{m}\right]$ and its tension is $20N$. The string passes through a bath filled with 1 kg water. Due to friction, heat is transferred to the bath. The heat transfer efficiency is 50%.

20. The power transfer to the bath is :-

(A) 0.0512 watt (B) 0.0124 watt (C) 0.1024 watt (D) 0.0345 watt

PWV239

21. The time passes before the temperature of the bath rises one degree kelvin is nearly :-

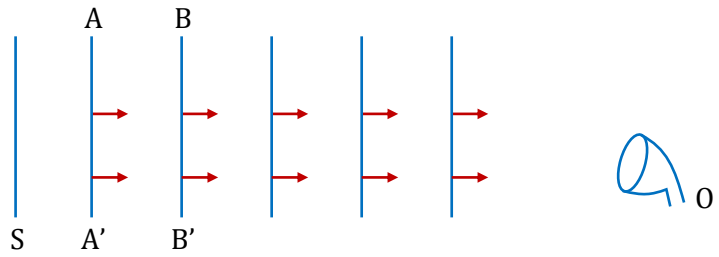
(A) 11.4 hr. (B) 22.8 hr. (C) 5.6 hr. (D) 8 hr.

PWV240

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MATRIX MATCH TYPE QUESTION

22. Consider a large plane diaphragm '*S* ' emitting sound and a detector '*O*'. The diagram shows plane wavefronts for the sound wave travelling in air towards right when source, observer and medium are at rest. *AA*' and *BB*' are fixed imaginary planes. Column-I describes about the motion of source, observer or medium and column-II describes various effects. Match them correctly.



	Column-I		**Column-II**
(A)	Source starts moving towards right.	(P)	Distance between any two wavefronts will increase.
(B)	Air starts moving towards right.	(Q)	Distance between any two wavefronts will decrease.
(C)	Observer and source both move towards left with same speed.	(R)	The time needed by sound to move from plane *AA*' to *BB*' will increase.
(D)	Source and medium (air) both move towards right with same speed.	(S)	The time needed by sound to move from plane *AA*' to *BB*' will decrease.
		(T)	Frequency received by observer increases.

PWV225

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EXERCISE (S)

Equation of Travelling Wave

1. A plane progressive wave of frequency 25 Hz, amplitude 2.5×10^{-5} m and initial phase zero propagates along the negative x-direction with a velocity of 300 m/s. At any instant, the phase difference between the oscillations at two points $6m$ apart along the line of propagation is _____, and the corresponding amplitude difference is _____m.

PWV091

2. The wave function for a travelling wave on a string is given as

$y(x, t) = (0.350\ m) \sin\left(10\pi t - 3\pi x + \frac{\pi}{4}\right)$

(a) What are the speed and direction of travel of the wave?
(b) What is the vertical displacement of the string at $t = 0$, $x = 0.1$ m?
(c) What are wavelength and frequency of the wave?

PWV344

3. Show that the wave function $y = e^{b(x - vt)}$ is a solution of the linear wave equation.

PWV345

Speed of Travelling Wave

4. A long uniform string of mass density 0.1 kg/m is stretched with a force of 40 N. One end of the string ($x = 0$) is oscillating transversely (sinusoidally) with an amplitude of 0.02 m and a period of 0.1 sec, so that travelling waves in the $+x$ direction are set up.

(a) What is the velocity of the waves?
(b) What is their wavelength?
(c) If at the driving end ($x = 0$) the displacement (y) at $t = 0$ is 0.01 m with dy/dt negative, what is the equation of the travelling waves?

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PWV104

5. The extension in a string, obeying Hooke's law is x. The speed of wave in the stretched string is v. If the extension in the string is increased to $1.5\,x$, find the new speed of wave.

PWV106

6. A non-uniform rope of mass M and length L has a variable linear mass density given by $\mu = kx$ where x is the distance from one end of the wire and k is a constant.

(a) Show that $M = kL^2/2$
(b) Show that the time required for a pulse generated at one end of the wire to travel to the other end is given by $t = \sqrt{\frac{8ML}{9F}}$ where F (constant)is the tension in the wire.

PWV111

Energy and Power in Travelling Wave

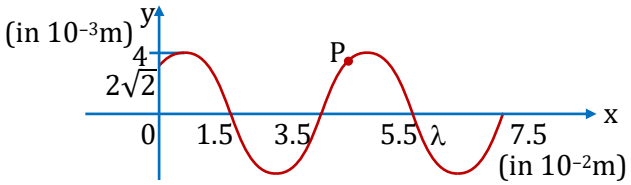
7. A long string under tension of 100 N has one end at $x = 0$. A sinusoidal wave is generated at $x = 0$ whose equation is given by $y = (0.01\ cm) \sin\left[\left(\frac{\pi x}{10} m\right) - 50\pi t(\text{sec})\right]$

(i) Sketch the shape of the string at $t = \frac{1}{50}$ sec.
(ii) Find the average power transmitted by the wave.
(iii) Draw velocity time graph of particle at $x = 5$ m.

PWV143

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8. The figure shows a snap photograph of a vibrating string at $t = 0$. The particle P is observed moving up with velocity 20π cm/s. The angle made by string with x-axis at P is 6°.



(a) Find the direction in which the wave is moving.

(b) The equation of the wave.

(c) The total energy carried by the wave per cycle of the string, assuming that μ, the mass per unit length of the string = 50 gm/m.

PWV144

9. Sinusoidal waves 5 mm in amplitude are to be transmitted along a string having a linear mass density equal to 4×10^{-2} kg/m. If the source can deliver an average power of 90 W and the string is under a tension of 100 N, then find the frequency (in Hz) at which the source can operate.
(Take $\pi^2 = 10$)

PWV147

10. A progressive wave on a string having linear mass density ρ is represented by $y = A\sin\left(\frac{2\pi}{\lambda}x - \omega t\right)$

where y is in mm. Find the total energy (in μJ) passing through origin from $t = 0$ to $t = \frac{\pi}{2\omega}$.

[Take : $\rho = 3 \times 10^{-2}$ kg/m; $A = 1mm$; $\omega = 100$ rad/sec; $\lambda = 16$ cm]

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PWV149

Standing Waves

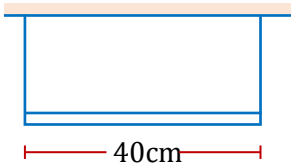
11. A 40 cm long wire having a mass 3.2 gm and area of cross-section 1 mm^2 is stretched between the supports 40.05 cm apart. In its fundamental mode it vibrates with a frequency of 1000/64 Hz. Find the young's modulus of the wire.

PWV131

12. A string, fixed at both ends, vibrates in a resonant mode with a separation of 2.0 cm between the consecutive nodes. For the next higher resonant frequency, this separation is reduced to 1.6 cm. Find the length of the string.

PWV136

13. A uniform horizontal rod of length 40 cm and mass 1.2 kg is supported by two identical wires as shown in figure. Where should a mass of 4.8 kg be placed on the rod so that the same tuning fork may excite the wire on left into its fundamental vibrations and that on right into its first overtone? Take $g = 10$ ms^{-2}.



PWV138

For More Join: @IITJEE_Advanced

14. Figure shows an aluminium wire of length 60 *cm* joined to a steel wire of length 80 *cm* and stretched between two fixed supports. The tension produced is 40 *N*. The cross-sectional area of the steel wire is 1.0 mm^2 and that of the aluminium wire is 3.0 mm^2. What could be the minimum frequency of a tuning fork which can produce standing waves in the system with the joint as a node? The density of aluminium is 2.6 $g\ cm^{-3}$ and that of steel is 7.8 $g\ cm^{-3}$.



PWV139

15. A particular guitar wire is 30.0 *cm* long and vibrates at a frequency of 196 *Hz* in fundamental mode when no finger is placed on it. The next higher notes on the scale are 220 *Hz*, 247 *Hz*, 262 *Hz* and 294 *Hz*. How far from the end of the string must the finger be placed to play these notes ?

PWV246

Properties of Sound

16. The sound level at a point 5.0 *m* away from a point source is 40*dB*. What will be the level at a point 50 *m* away from the source?

PWV172

17. Sound with intensity larger than 120*dB* appears painful to a person. A small speaker delivers 2.0 *W* of audio output. How close can the person get to the speaker without hurting his ears?

PWV174

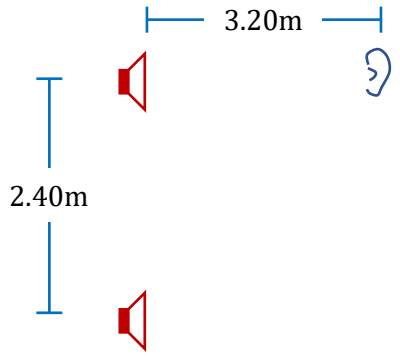
Rankers Academy JEE

Interference of Waves

18. In Quincke's experiment the sound detected is changed from a maximum to a minimum when the sliding tube is moved through a distance of 2.50 *cm*. Find the frequency of sound if the speed of sound in air is 340 ms^{-1}.

PWV180

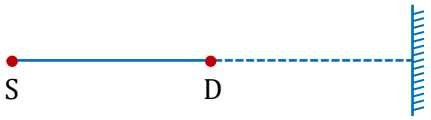
19. Two stereo speakers are separated by a distance of 2.40*m*. A person stands at a distance of 3.20 *m* directly in front of one of the speakers as shown in figure. Find the frequencies in the audible range (20-2000 *Hz*) for which the listener will hear a minimum sound intensity. Speed of sound in air = 320 ms^{-1}.



PWV181

For More Join: @IITJEE_Advanced

20. A source of sound S and a detector D are placed at some distance from one another. A big cardboard is placed near the detector and perpendicular to the line SD as shown in figure. It is gradually moved away and it is found that the intensity changes from a maximum to a minimum as the board is moved through a distance of 20 cm. Find the frequency of the sound emitted. Velocity of sound in air is 336 ms^{-1}.



PWV182

21. Two sources of sound S_1 and S_2, emitting waves of equal wavelength 20.0 cm, are placed with a separation of 20.0 cm between them. A detector can be moved on a line parallel to S_1 S_2 and at a distance of 20.0 cm from it. Initially, the detector is equidistant from the two sources. Assuming that the waves emitted by the sources are in phase, find the minimum distance through which the detector should be shifted to detect a minimum of sound.

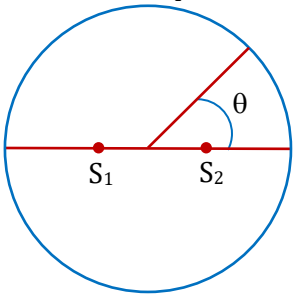
PWV183

22. Three sources of sound S_1, S_2 and S_3 of equal intensity are placed in a straight line with $S_1S_2 = S_2S_3$. At a point P, far away from the sources, the wave coming from S_2 is 120° ahead in phase of that from S_1. Also, the wave coming from S_3 is 120° ahead of that from S_2. What would be the resultant intensity of sound at P?



PWV184

23. Figure shows two coherent sources S_1 and S_2 which emit sound of wavelength λ in phase. The separation between the sources is 3λ. A circular wire of large radius is placed in such a way that S_1 and S_2 lies in its plane and the middle point of S_1, S_2 is at the centre of the wire. Find the value of $\cos\theta$ for which constructive interference takes place.



PWV185

Standing Wave & Beats

24. In a closed organ pipe, the frequency of fundamental note is 30 Hz. A certain amount of water is now poured in the organ pipe so that the fundamental frequency is increased to 110 Hz. If the organ pipe has a cross-sectional area of 2 cm^2, the amount of water poured in the organ tube is _____ g. (Take speed of sound in air is 330 m/s)

PWV281

25. The fundamental frequency of a closed organ pipe is equal to the first overtone frequency of an open organ pipe. If length of the open pipe is 60 cm, the length of the closed pipe will be:

PWV282

For More Join: @IITJEE_Advanced

26. Two narrow cylindrical pipes A and B have the same length. Pipe A is open at both ends and is filled with a monoatomic gas of molar mass M_A. Pipe B is open at one end and closed at the other end, and is filled with a diatomic gas of molar mass M_B. Both gases are at the same temperature.

(a) If the frequency of the second harmonic of the fundamental mode in pipe A is equal to the frequency of the third harmonic of the fundamental mode in pipe B, determine the value of M_A/M_B.

(b) Now the open end of pipe B is also closed (so that the pipe is closed at both ends). Find the ratio of the fundamental frequency in pipe A to that in pipe B.

PWV193

27. A Kundt's tube apparatus has a copper rod of length 1.0 m clamped at 25 cm from one of the ends. The tube contains air in which the speed of sound is 340 m/s. the powder collects in heaps separated by a distance of 5.0 cm. Find the speed of sound waves in copper.

PWV207

28. A source of sound with adjustable frequency produces 2 beats per second with a tuning fork when its frequency is either 476 Hz or 480 Hz. What is the frequency of the tuning fork?

PWV208

29. A tuning fork of frequency 256 Hz produces 4 beats per second with a wire of length 25 cm vibrating in its fundamental mode. The beat frequency decreases when the length is slightly shortened. What could be the minimum length by which the wire be shortened so that it produces no beats with the tuning fork?

PWV210

30. A, B and C are three tuning forks. Frequency of A is 350 Hz. Beats produced by A and B are 5 per second and by B and C are 4 per second. When a wax is put on A beat frequency between A and B is 2Hz and between A and C is 6Hz. Then, find the frequency of B and C respectively.

Kankers' Academy JEE

PWV211

Doppler Effect

31. A boat is travelling in a river with a speed of 10 m/s along the stream flowing with a speed 2 m/s. From this boat, a sound transmitter is lowered into the river through a rigid support. The wavelength of the sound emitted from the transmitter inside the water is 14.45 mm. Assume that attenuation of sound in water and air is negligible.

(a) What will be the frequency detected by a receiver kept inside the river downstream ?

(b) The transmitter and the receiver are now pulled up into air. The air is blowing with a speed 5 m/sec in the direction opposite the river stream. Determine the frequency of the sound detected by the receiver.

(Temperature of the air and water = 20°C ; Density of river water = 10^3 Kg/m^3; Bulk modulus of the water = 2.088 × 10^9 Pa; Gas constant R = 8.31 $J/mol\text{-}K$; Mean molecular mass of air = 28.8 × 10^{-3} kg/mol; C_P/C_V for air = 1.4)

Note: Boat velocity is with respect to ground and receiver is stationary w.r.t. ground.

PWV230

32. A supersonic jet plane moves parallel to the ground at speed v = 0.75 mach (1 mach = speed of sound). The frequency of its engine sound is ν_0 = 2 kHz and the height of the jet plane is h = 1.5 km. At some instant an observer on the ground hears a sound of frequency ν = 2 ν_0. Find the instant prior to the instant of hearing when the sound wave received by the observer was emitted by the jet plane. Velocity of sound wave in the condition of observer = 340 m/s.

PWV250

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. A train is moving on a straight track with speed 20 ms^{-1}. It is blowing its whistle at the frequency of 1000 Hz. The percentage change in the frequency heard by a person standing near the track as the train passes him is (speed of sound = 320 ms^{-1}) close to: **[JEE (Main) 2015]**

(1) 6% (2) 12% (3) 18% (4) 24%

PWV033

2. An observer is moving with half the speed of light towards stationary microwave source emitting waves at frequency 10GHz. What is the frequency of the microwave measured by the observer? (speed of light = 3×10^8 ms^{-1}) **[JEE (Main) 2017]**

(1) 15.3 GHz (2) 10.1 GHz (3) 12.1 GHz (4) 17.3 GHz

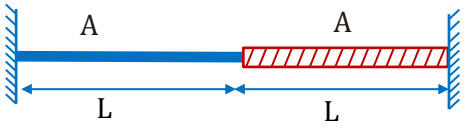
PWV034

3. A heavy ball of mass M is suspended from the ceiling of a car by a light string of mass m ($m << M$). When the car is at rest, the speed of transverse waves in the string is 60 ms^{-1}. When the car has acceleration a, the wave-speed increases to 60.5 ms^{-1}. The value of a, in terms of gravitational acceleration g, is closest to: **[JEE (Main) 2019]**

(1) $\frac{g}{5}$ (2) $\frac{g}{20}$ (3) $\frac{g}{10}$ (4) $\frac{g}{30}$

PWV035

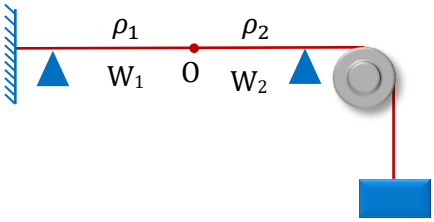
4. A wire of length 2L, is made by joining two wires A and B of same length but different radii r and 2r and made of the same material. It is vibrating at a frequency such that the joint of the two wires forms a node. If the number of antinodes in wire A is p and that in B is q then the ratio $p : q$ is : **[JEE (Main) 2019]**



(1) 4 : 9 (2) 3 : 5 (3) 1 : 4 (4) 1 : 2

PWV036

5. Two wires W_1 and W_2 have the same radius r and respective densities ρ_1 and ρ_2 such that $\rho_2 = 4\rho_1$. They are joined together at the point O, as shown in the figure. The combination is used as a sonometer wire and kept under tension T. The point O is midway between the two bridges. When a stationary wave is set up in the composite wire, the joint is found to be a node. The ratio of the number of antinodes formed in W_1 to W_2 is : **[JEE (Main) 2019]**



(1) 1 : 3 (2) 1 : 1 (3) 4 : 1 (4) 1 : 2

PWV038

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6. A person standing on an open ground hears the sound of a jet Aeroplane, coming from north at an angle 60° with ground level. But he finds the Aeroplane right vertically above his position. If v is the speed of sound, speed of the plane is: **[JEE (Main) 2019]**

(1) $\frac{2v}{\sqrt{3}}$ (2) v (3) $\frac{v}{2}$ (4) $\frac{\sqrt{3}}{2}v$

PWV040

7. In a resonance tube experiment when the tube is filled with water up to height of 17.0 *cm* from bottom, it resonates with a given tuning fork. When the water level is raised the next resonance with the same tuning fork occurs at a height of 24.5 *cm*. If the velocity of sound in air is 330 m/s, the tuning fork frequency is: **[JEE (Main) 2020]**

(1) 1100 *Hz* (2) 3300 *Hz* (3) 2200 *Hz* (4) 550 *Hz*

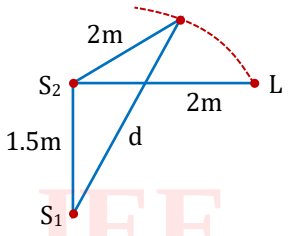
PWV041

8. Assume that the displacement(s) of air is proportional to the pressure difference (Δp) created by a sound wave. Displacement(s) further depends on the speed of sound (v), density of air (ρ) and the frequency (f). If $\Delta p \sim 10\ Pa$, $v \sim 300\ m/s$, $\rho \sim 1\ kg/m^3$ and $f \sim 1000\ Hz$, then s will be the order of (Take multiplicative constant to be 1) **[JEE (Main) 2020]**

(1) 10 *mm* (2) $\frac{3}{100}$ *mm* (3) 1 *mm* (4) $\frac{1}{10}$ *mm*

PWV042

9. Two coherent sources of sound, S_1 and S_2, produce sound waves of the same wavelength, $\lambda = 1\ m$, in phase. S_1 and S_2 are placed 1.5 *m* apart (see fig.) A listener, located at L, directly in front of S_2 finds that the intensity is at a minimum when he is $2m$ away from S_2. The listener moves away from S_1, keeping his distance from S_2 fixed. The adjacent maximum of intensity is observed when the listener is at a distance d from S_1. Then, d is :



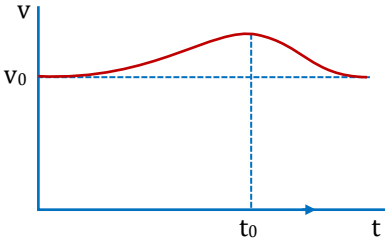
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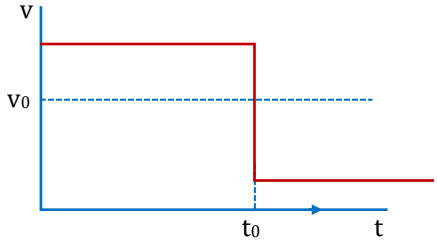
[JEE (Main) 2020]

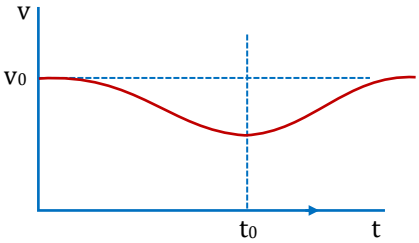
(1) $12m$ (2) $3m$ (3) $5m$ (4) $2m$

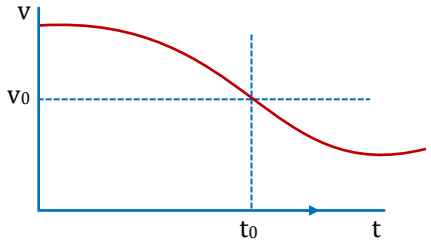
PWV043

10. A sound source S is moving along a straight track with speed v, and is emitting sound of frequency v_0. An observer is standing at a finite distance, at the point O, from the track. The time variation of frequency heard by the observer is best represented by:
(t_0 represents the instant when the distance between the source and observer is minimum)
[JEE (Main) 2020]

(1)


(2)


(3)


(4)


PWV044

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11. Two waves are simultaneously passing through a string and their equations are : $y_1 = A_1 \sin k(x - vt)$, $y_2 = A_2 \sin k(x - vt + x_0)$. Given amplitudes $A_1 = 12\ mm$ and $A_2 = 5\ mm$, $x_0 = 3.5\ cm$ and wave number $k = 6.28\ cm^{-1}$. The amplitude of resulting wave will be _____ mm.

[JEE (Main) 2021]

PWV045

12. Sound travels in a mixture of two moles of helium and n moles of hydrogen. If rms speed of gas molecules in the mixture is $\sqrt{2}$ times the speed of sound, then the value of n will be

[JEE (Main) 2022]

(1) 1 (2) 2 (3) 3 (4) 4

PWV047

13. A transverse wave is represented by $y = 2 \sin(\omega t - kx)$ cm. The value of wavelength (in cm) for which the wave velocity becomes equal to the maximum particle velocity, will be

[JEE (Main) 2022]

(1) 4π (2) 2π (3) π (4) 2

PWV048

14. A person observes two moving trains, 'A' reaching the station and 'B' leaving the station with equal speed of 30 m/s. If both trains emit sounds with frequency 300 Hz, (Speed of sound : 330 m/s) approximate difference of frequencies heard by the person will be : **[JEE (Main) 2023]**

(1) 33 Hz (2) 55 Hz (3) 80 Hz (4) 10 Hz

PWV050

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15. A closed organ pipe 150 cm long gives 7 beats per second with an open organ pipe of length 350 cm, both vibrating in fundamental mode. The velocity of sound is _____ m/s.

[JEE (Main) 2024]

PWV283

16. Two waves of intensity ratio 1 : 9 cross each other at a point. The resultant intensities at the point, when (a) Waves are incoherent is I_1 (b) Waves are coherent is I_2 and differ in phase by 60°. If $\frac{I_1}{I_2} = \frac{10}{x}$ then x = _____.

[JEE (Main) 2024]

PWV284

17. The equation of stationary wave is :

$$y = 2a \sin\left(\frac{2\pi nt}{\lambda}\right) \cos\left(\frac{2\pi x}{\lambda}\right)$$

Which of the following is NOT correct? **[JEE (Main) 2024]**

(1) The dimensions of nt is $[L]$ (2) The dimensions of n is $[LT^{-1}]$

(3) The dimensions of n/λ is $[T]$ (4) The dimensions of x is $[L]$

PWV285

18. A sonometer wire of resonating length 90 cm has a fundamental frequency of 400 Hz when kept under some tension. The resonating length of the wire with fundamental frequency of 600 Hz under same tension _____ cm. **[JEE (Main) 2024]**

PWV286

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19. The equation of a transverse wave travelling along a string is $y(x, t) = 4.0 \sin [20 \times 10^{-3} x + 600t]$ *mm*, where x is in the *mm* and t is in second. The velocity of the wave is : **[JEE (Main) 2025]**

(1) + 30 m/s (2) – 60 m/s (3) – 30 m/s (4) + 60 m/s

PWV339

20. A closed organ and an open organ tube filled by two different gases having same bulk modulus but different densities ρ_1 and ρ_2 respectively. The frequency of 9th harmonic of closed tube is identical with 4th harmonic of open tube. If the length of the closed tube is 10 *cm* and the density ratio of the gases is $\rho_1 : \rho_2 = 1 : 16$, then the length of the open tube is : **[JEE (Main) 2025]**

(1) $\frac{20}{7} cm$ (2) $\frac{15}{7} cm$ (3) $\frac{20}{9} cm$ (4) $\frac{15}{9} cm$

PWV340

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EXERCISE - JEE (Advanced) PYQ

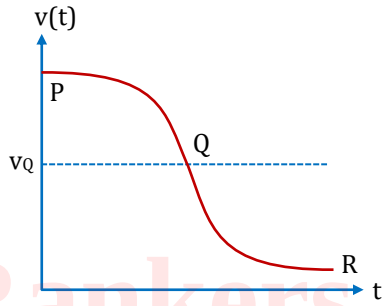
1. Four harmonic waves of equal frequencies and equal intensities I_0 have phase angles 0, $\pi/3$, $2\pi/3$ and π. When they are superposed, the intensity of the resulting wave is nI_0. The value of n is.

[JEE (Advanced) 2015]

PWV073

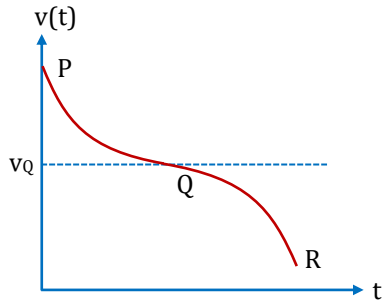
2. Two loudspeakers M and N are located $20m$ apart and emit sound at frequencies 118 Hz and 121 Hz, respectively. A car is initially at a point P, 1800 m away from the midpoint Q of the line MN and moves towards Q constantly at 60 km/hr along the perpendicular bisector of MN. It crosses Q and eventually reaches a point R, 1800 m away from Q. Let $\nu(t)$ represent the beat frequency measured by a person sitting in the car at time t. Let v_P, v_Q and v_R be the beat frequencies measured at locations P, Q and R, respectively. The speed of sound in air is 330 ms^{-1}. Which of the following statement(s) is(are) true regarding the sound heard by the person? **[JEE (Advanced) 2016]**

(A) The plot below represents schematically the variation of beat frequency with time.



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(B) The plot below represents schematically the variations of beat frequency with time.

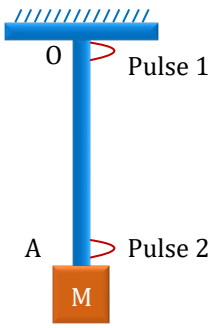


(C) The rate of change in beat frequency is maximum when the car passes through Q.

(D) $v_P + v_R = 2v_Q$

PWV074

3. A block M hangs vertically at the bottom end of a uniform rope of constant mass per unit length. The top end of the rope is attached to a fixed rigid support at O. A transverse wave pulse (Pulse 1) of wavelength λ_0 is produced at point O on the rope. The pulse takes time T_{OA} to reach point A. If the wave pulse of wavelength λ_0 is produced at point A (Pulse 2) without disturbing the position of M it takes time T_{AO} to reach point O. Which of the following options is/are correct



[JEE (Advanced) 2017]

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(A) The time $T_{AO} = T_{OA}$
(B) The velocities of the two pulses (Pulse 1 and Pulse 2) are the same at the midpoint of rope
(C) The wavelength of Pulse 1 becomes longer when it reaches point A
(D) The velocity of any pulse along the rope is independent of its frequency and wavelength.

PWV070

4. A stationary source emits sound of frequency $f_0 = 492\ Hz$. The sound is reflected by a large car approaching the source with a speed of $2\ ms^{-1}$. The reflected signal is received by the source and superposed with the original. What will be the beat frequency of the resulting signal in Hz ?
(Given that the speed of sound in air is $330\ ms^{-1}$ and the car reflects the sound at the frequency it has received). **[JEE (Advanced) 2017]**

PWV075

5. Two men are walking along a horizontal straight line in the same direction. The man in front walks at a speed $1.0\ ms^{-1}$ and the man behind walks at a speed $2.0\ ms^{-1}$. A third man is standing at a height $12m$ above the same horizontal line such that all three men are in a vertical plane. The two walking men are blowing identical whistles which emit a sound of frequency $1430\ Hz$. The speed of sound in air is $330\ ms^{-1}$. At the instant, when the moving men are $10\ m$ apart, the stationary man is equidistant from them. The frequency of beats in Hz, heard by the stationary man at this instant, is _____. **[JEE (Advanced) 2018]**

PWV076

6. In an experiment to measure the speed of sound by a resonating air column, a tuning fork of frequency $500\ Hz$ is used. The length of the air column is varied by changing the level of water in the resonance tube. Two successive resonances are heard at air columns of length $50.7\ cm$ and $83.9\ cm$. Which of the following statements is (are) true ? **[JEE (Advanced) 2018]**
(A) The speed of sound determined from this experiment is $332\ ms^{-1}$.
(B) The end correction in this experiment is $0.9\ cm$.
(C) The wavelength of the sound wave is $66.4\ cm$.
(D) The resonance at $50.7\ cm$ corresponds to the fundamental harmonic.

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PWV077

7. Answer the following by appropriately matching the lists based on the information given in the paragraph. A musical instrument is made using four different metal strings, 1, 2, 3 and 4 with mass per unit length $\mu, 2\mu, 3\mu$ and 4μ respectively. The instrument is played by vibrating the strings by varying the free length in between the range L_0 and $2L_0$. It is found that in string-1 (μ) at free length L_0 and tension T_0 the fundamental mode frequency is f_0. **[JEE (Advanced) 2019]**
List-I gives the above four strings while list-II lists the magnitude of some quantity.

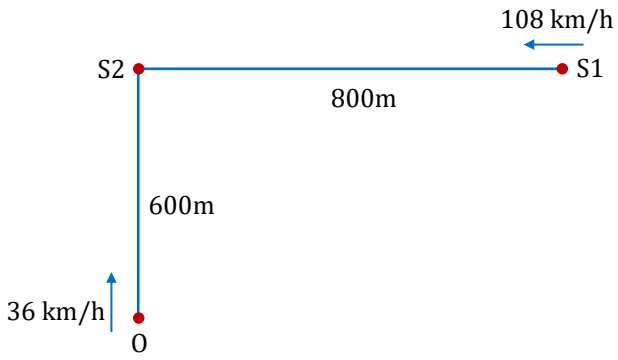
	List-I		**List-II**
(I)	String-1 (μ)	(P)	1
(II)	String-2 (2μ)	(Q)	$1/2$
(III)	String-3 (3μ)	(R)	$1/\sqrt{2}$
(IV)	String-4 (4μ)	(S)	$1/\sqrt{3}$
		(T)	$3/16$
		(U)	$1/16$

The length of the string 1,2,3 and 4 are kept fixed at $L_0, \frac{3L_0}{2}, \frac{5L_0}{4}$ and $\frac{7L_0}{4}$, respectively. Strings 1, 2, 3 and 4 are vibrated at their 1st , 3rd, 5th and 14th harmonics, respectively such that all the strings have same frequency. The correct match for the tension in the four strings in the units of T_0 will be.
(A) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ T, IV $\rightarrow$ U
(B) I $\rightarrow$ T, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ U
(C) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ T
(D) I $\rightarrow$ P, II $\rightarrow$ R, III $\rightarrow$ T, IV $\rightarrow$ U

PWV071

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8. A train S_1, moving with a uniform velocity of 108 km/h, approaches another train S_2 standing on a platform. An observer O moves with a uniform velocity of 36 km/h towards S_2, as shown in figure. Both the trains are blowing whistles of same frequency 120 Hz. When O is 600 m away from S_2 and distance between S_1 and S_2 is 800 m, the number of beats heard by O is _____.
[Speed of the sound = 330 m/s] **[JEE (Advanced) 2019]**



PWV078

9. A stationary tuning fork is in resonance with an air column in a pipe. If the tuning fork is moved with a speed of 2 ms^{-1} in front of the open end of the pipe and parallel to it, the length of the pipe should be changed for the resonance to occur with the moving tuning fork. If the speed of sound in air is 320 ms^{-1}, the smallest value of the percentage change required in the length of the pipe is _____. **[JEE (Advanced) 2020]**

PWV079

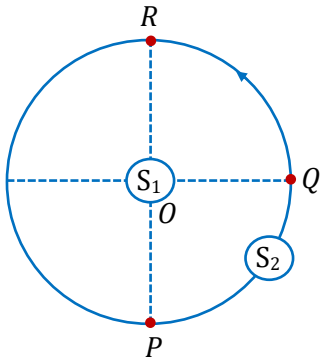
10. A source, approaching with speed u towards the open end of a stationary pipe of length L, is emitting a sound of frequency f_s. The farther end of the pipe is closed. The speed of sound in air is v and f_0 is the fundamental frequency of the pipe. For which of the following combination(s) of u and f_s, will the sound reaching the pipe lead to a resonance ? **[JEE (Advanced) 2021]**

(A) $u = 0.8\ v$ and $f_s = f_0$ (B) $u = 0.8\ v$ and $f_s = 2f_0$
(C) $u = 0.8\ v$ and $f_s = 0.5\ f_0$ (D) $u = 0.5\ v$ and $f_s = 1.5\ f_0$

PWV080

PARAGRAPH FOR QUESTION NO. 11 AND 12

S_1 and S_2 are two identical sound sources of frequency 656 Hz. *The* source S_1 is located at O and S_2 moves anti-clockwise with a uniform speed $4\sqrt{2}\ ms^{-1}$ on a circular path around O, as shown in the figure. There are three points P, Q and R on this path such that P and R are diametrically opposite while Q is equidistant from them. A sound detector is placed at point P. The source S_1 can move along direction OP.
[Given: The speed of sound in air is 324 ms^{-1}] **[JEE (Advanced) 2023]**



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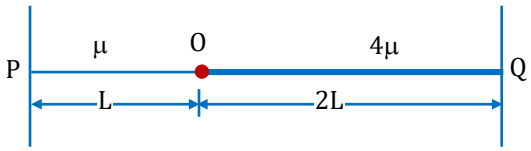
11. When only S_2 is emitting sound and it is Q, the frequency of sound measured by the detector in Hz is _____.

PWV259

12. Consider both sources emitting sound. When S_2 is at R and S_1 approaches the detector with a speed 4 ms^{-1}, the beat frequency measured by the detector is _____Hz.

PWV260

13. Two uniform string of mass per unit length μ and 4μ, and length L and $2L$, respectively, are joined at point O, and tied at two fixed ends P and Q, as shown in the figure. The strings are under a uniform tension T. If we define the frequency $v_0 = \frac{1}{2L}\sqrt{\frac{T}{\mu}}$, which of the following statement(s) is (are) correct ? **[JEE (Advanced) 2024]**



(A) With a node at O, the minimum frequency of vibration of the composite string is v_0.

(B) With an antinode at O, the minimum frequency of vibration of the composite string is $2v_0$.

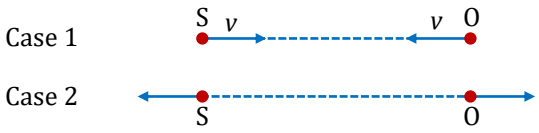
(C) When the composite string vibrates at the minimum frequency with a node at O, it has 6 nodes, including the end nodes.

(D) No vibrational mode with an antinode at O is possible for the composite string.

PWV287

14. A source (S) of sound has frequency 240 Hz. When the observer (O) and the source move towards each other at a speed v with respect to the ground (as shown in Case 1 in the figure), the observer measures the frequency of the sound to be 288 Hz. However, when the observer and the source move away from each other at the same speed v with respect to the ground (as shown in Case 2 in the figure), the observer measures the frequency of sound to be n Hz. The value of n is _____.

[JEE (Advanced) 2024]



PWV288

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	D	D	A	C	A	D	B	C	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	A	A	B	B	C	B	B	A	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	B	C	A	B	A	A	A	A	C
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	D	B	B	A	B	A	D	B	D
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	B	A	A	C	B	B	B	B	C
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	B	A	D	A	C	B	A	D	B	D
Que.	61	62	63	64						
Ans.	A	B	B	C						

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	ABCD	ACD	ABD	AC	BCD	ABD	BD	BD	AC	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	B	A	B	B	A	B	C	D	A
Que.	21	22								
Ans.	B	(A-Q, T), (B-P, S), (C-P), (D-S, T)								

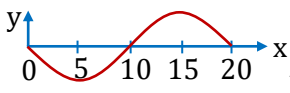
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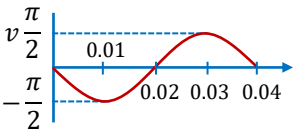
EXERCISE (S)

1. π, 0 **2.** (a) 3.33 *m/sec* and along *+ve x* axis, (b) – 5.48 *cm*, (c) 5 *Hz*.

3. Theoretical **4.** (a) 20 m/s; (b) 2 m; (c) $y(x, t) = 0.02\sin(\pi x - 20\pi t + \pi/6)$

5. 1.22 v **6.** (a) $M = \frac{\kappa L^2}{2}$ (b) $t = \sqrt{\frac{8\ ML}{9F}}$

7. (i) 

, (ii) $25 \times 10^{-6}\ W$, (iii) 

, solved

8. (a) Negative x ; (b) $y = 4 \times 10^{-3} \sin 100\pi\left(3t + 0.5x + \frac{1}{400}\right)$ (x , y in meter) ; (c) $144\ \pi^2 \times 10^{-5}\ J$

9. 300 **10.** 6 **11.** $1 \times 10^9\ Nm^2$ **12.** 8.0 *cm*

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13. 5 *cm* from the left end **14.** 180 *Hz*

15. 26.7 *cm*, 23.8 *cm*, 22.4 *cm* and 20.0 *cm* **16.** $\beta = 20dB$ **17.** 40 *cm*

18. 3.4 *kHz* **19.** $f = 200\,(2n + 1)$, where $n = 1, 2, 3\ldots, 49$ **20.** 420*Hz*

21. 12.6 *cm* **22.** 0 **23.** $0, \frac{1}{3}, \frac{2}{3}$ ..... and so on **24.** 400

25. 15 *cm* **26.** (a) 2.116, (b) $\frac{3}{4}$ **27.** 3400 m/s **28.** 478 *Hz*

29. 0.39 *cm* **30.** 345, 341 or 349 *Hz*

31. (a) 100696 *Hz* (b) 103038 *Hz* **32.** 5.9 sec

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	4	1	4	4	3	3	2	2	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	7	2	1	2	294	13	3	60	3	3

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8
Ans.	3	ACD	AD	6	5 [4.99, 5.01]	ABC	A	8.12 to 8.13
Que.	9		10	11	12	13	14	
Ans.	0.62 to 0.64		AD	648	8 to 8.40	ACD	200	

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